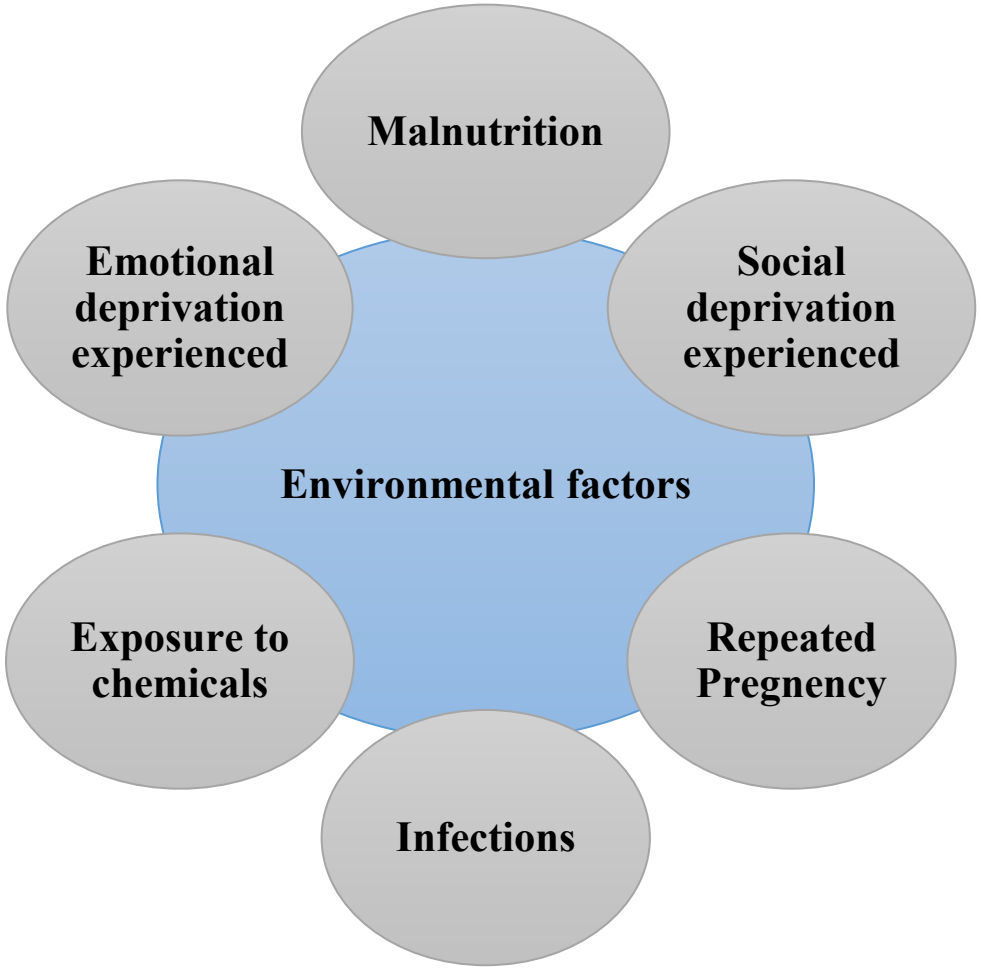
INVESTIGATING THE AUTOSOMAL RECESSIVE INHERITANCE PATTERN OF *MLC-1* GENE MUTATIONS IN NON-SYNDROMIC INTELLECTUAL DISABILITY PATIENTS

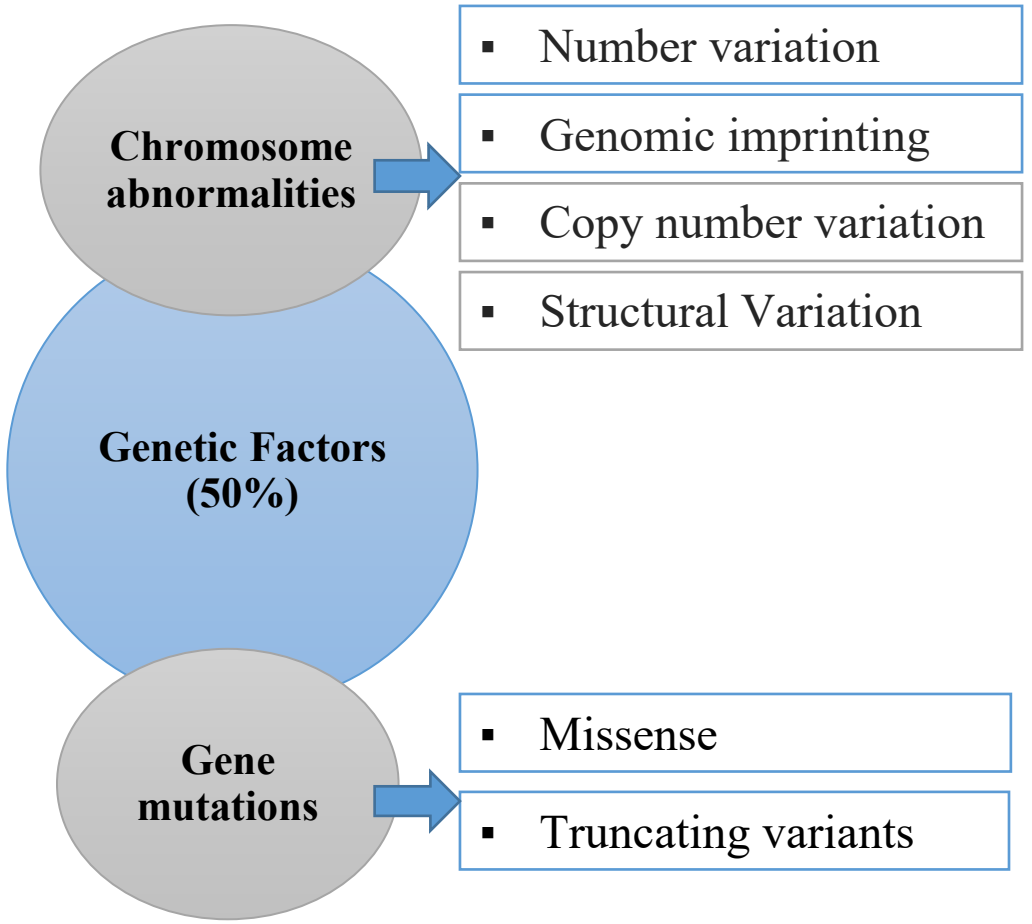
Azka Shahid Khan
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INTRODUCTION

- Intellectual disability (ID), also referred to as **mental retardation.**
- A condition characterized by significant limitation in both intellectual functioning and adaptive behavior.
- Diagnosed when IQ is below **70–75**.
- Onset occurs **before 18 years** of age.
- 1-3% of the global population with a male-to-female ratio of 2:1. (Maulik *et al*., 2011)

ETIOLOGY OF ID





CLASSIFICATION OF ID

Level of ID	IQ Score	Prevalence (% of total ID cases)	Adult Skills
Mild ID	50-70	46.7%	Requires minimal support
Moderate ID	35-50	32.1%	Limited academic achievement
Severe ID	20-35	14.6%	Limited verbal communication and mobility
Profound ID	Below 20	6.7%	Associated with multiple disabilities

(Naz *et al.*, 2021)

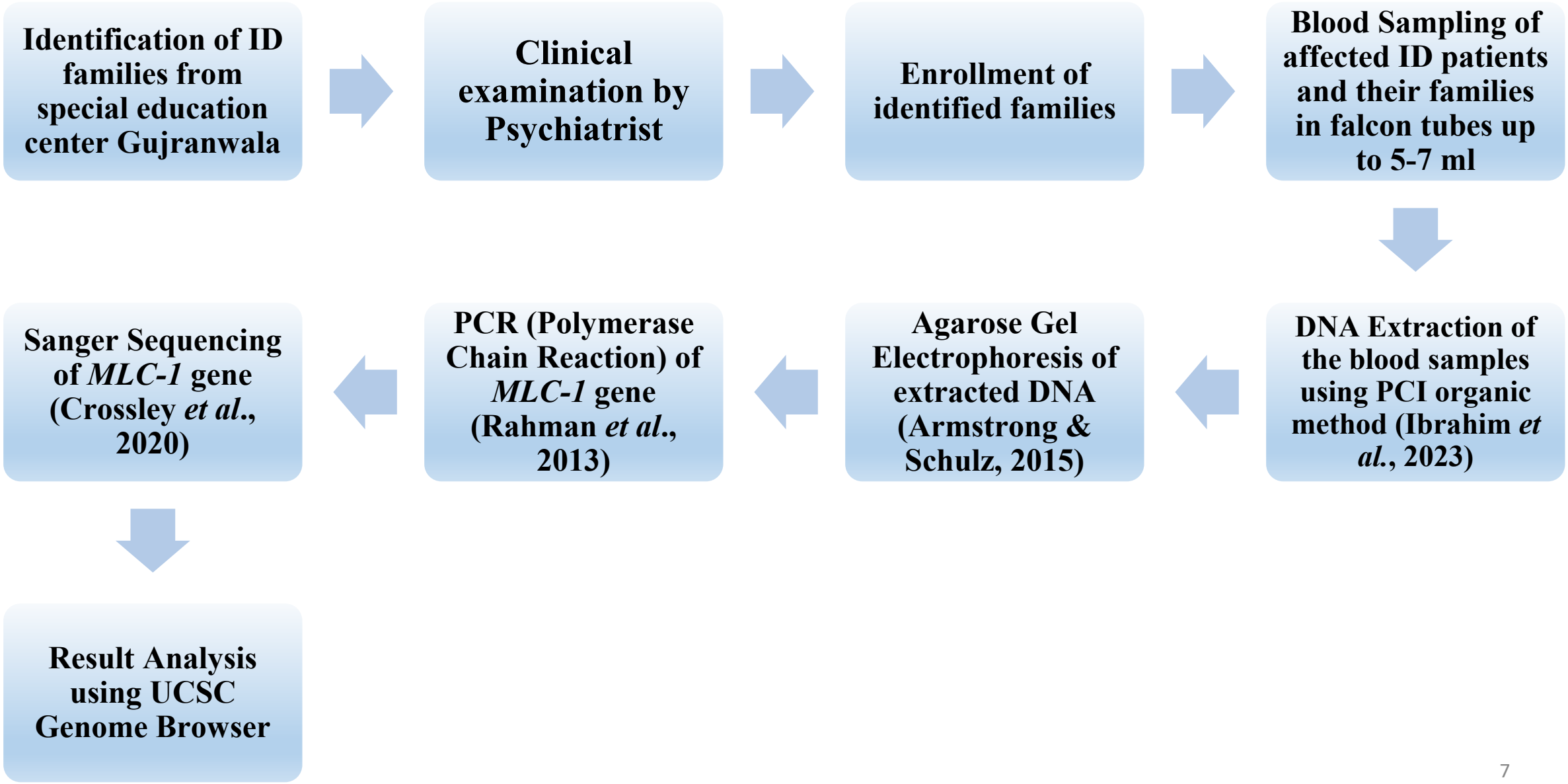
MLC-1 GENE

- Mutations in this gene can cause a **neurological** disorder known as megalencephalic leukoencephalopathy with subcortical cysts (MLC).
- Provides instructions for making a protein called MLC1 (Modulator of VRAC current 1)
- Located on chromosome **22q13.33**, comprises **12 exons**.
- Plays a critical role in maintaining **astrocyte function** and **brain homeostasis**
- Follow an **autosomal recessive inheritance pattern.**

AIM AND OBJECTIVES

- This study aimed to investigate the autosomal recessive inheritance pattern of *MLC-1* gene mutations in non-syndromic intellectual disability (NSID) patients.
- The objective of the study was to conduct genetic analyses of individuals with NSID and their families.

MATERIAL AND METHODS





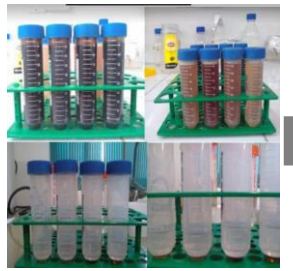
Thawing frozen blood for DNA extraction



Lysis and washing (Tris-HCl and EDTA buffer)



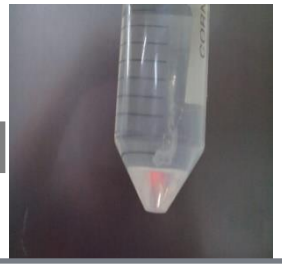
Removal of RBCs by centrifugation at 3000rpm



Repeat it for three times to remove RBCs



Resuspend pellets in TNE, proteinase K along with 10% SDS



NaCl to precipitate the proteins, Isopropanol centrifuge



Resuspend the pellet in TE buffer injection water for storing



Determination of DNA concentration by agarose gel electrophoresis



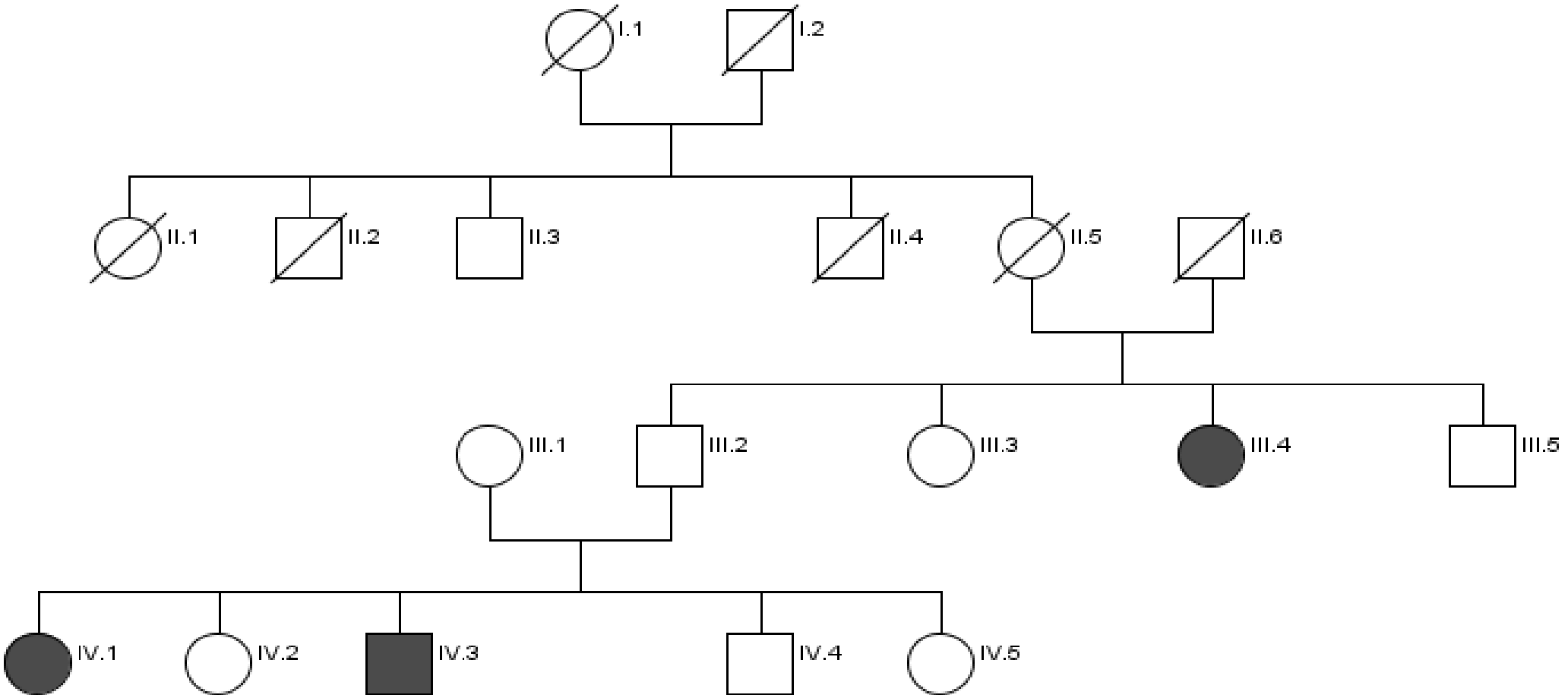
PCR(polymerase chain reaction)

Sanger sequencing was performed commercially

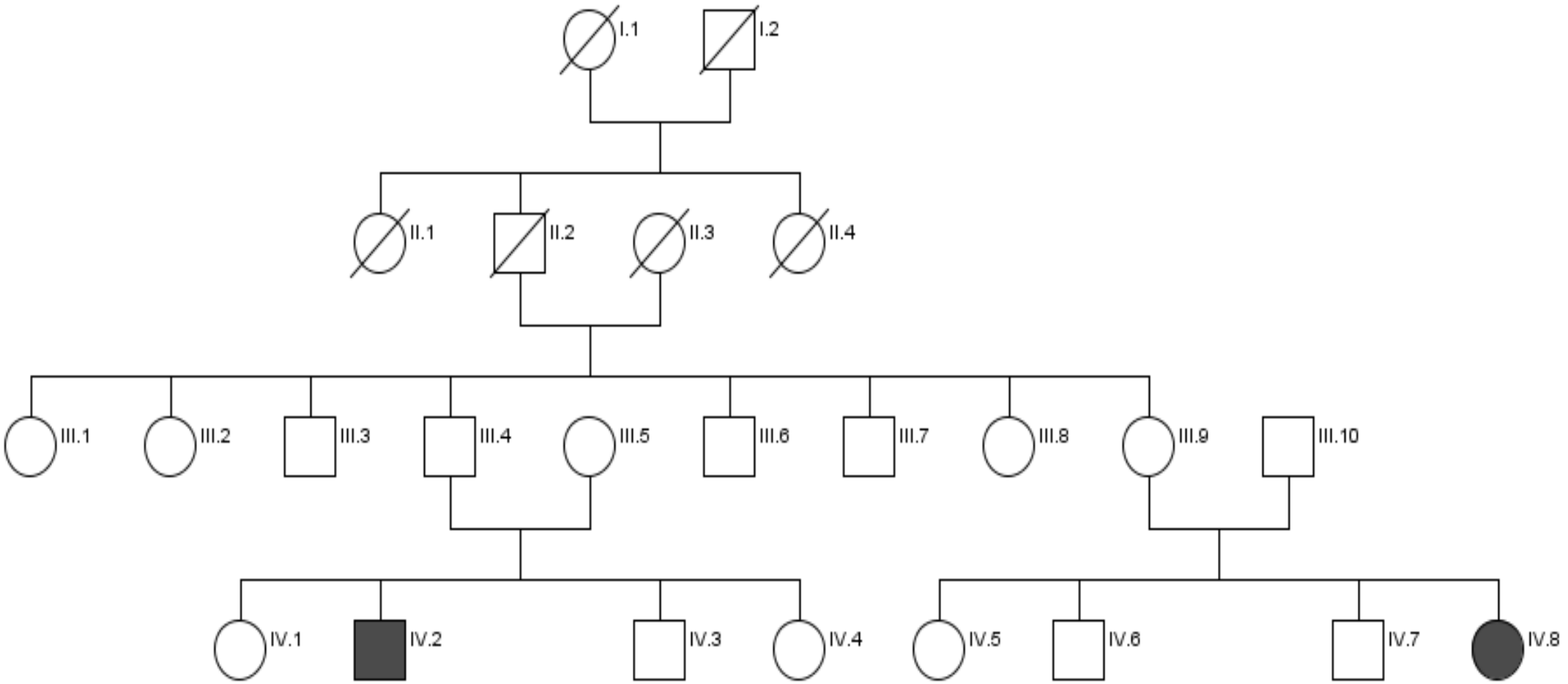
(Grimberg *et al.*,1989)

ID FAMILIES

- Two families PKID0139 and PKID0140 identified from Gujranwala.
- PKID0139 and PKID0140 both has two affected individuals.



Pedigree of family PKID0139



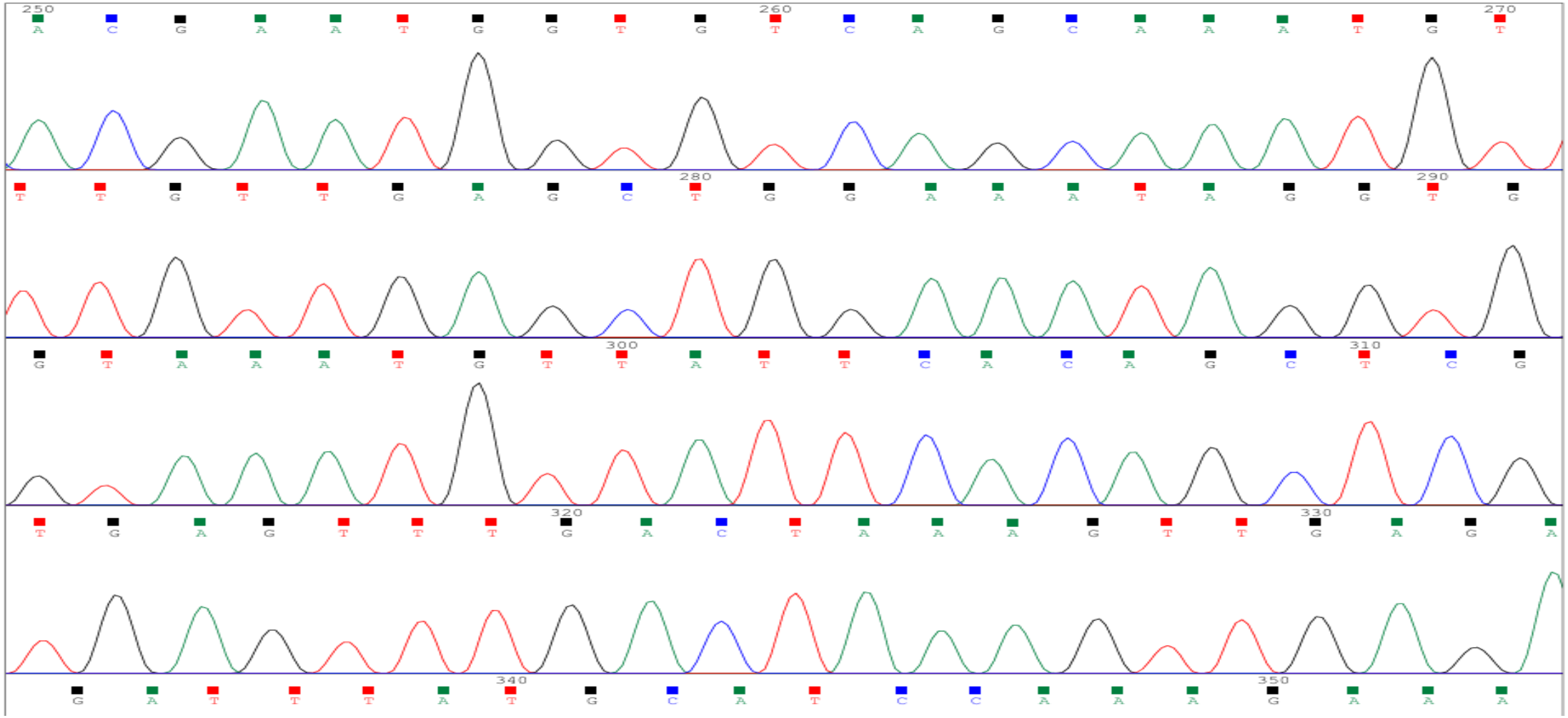
Pedigree of family PKID0140

ANALYSIS OF SEQUENCING RESULTS

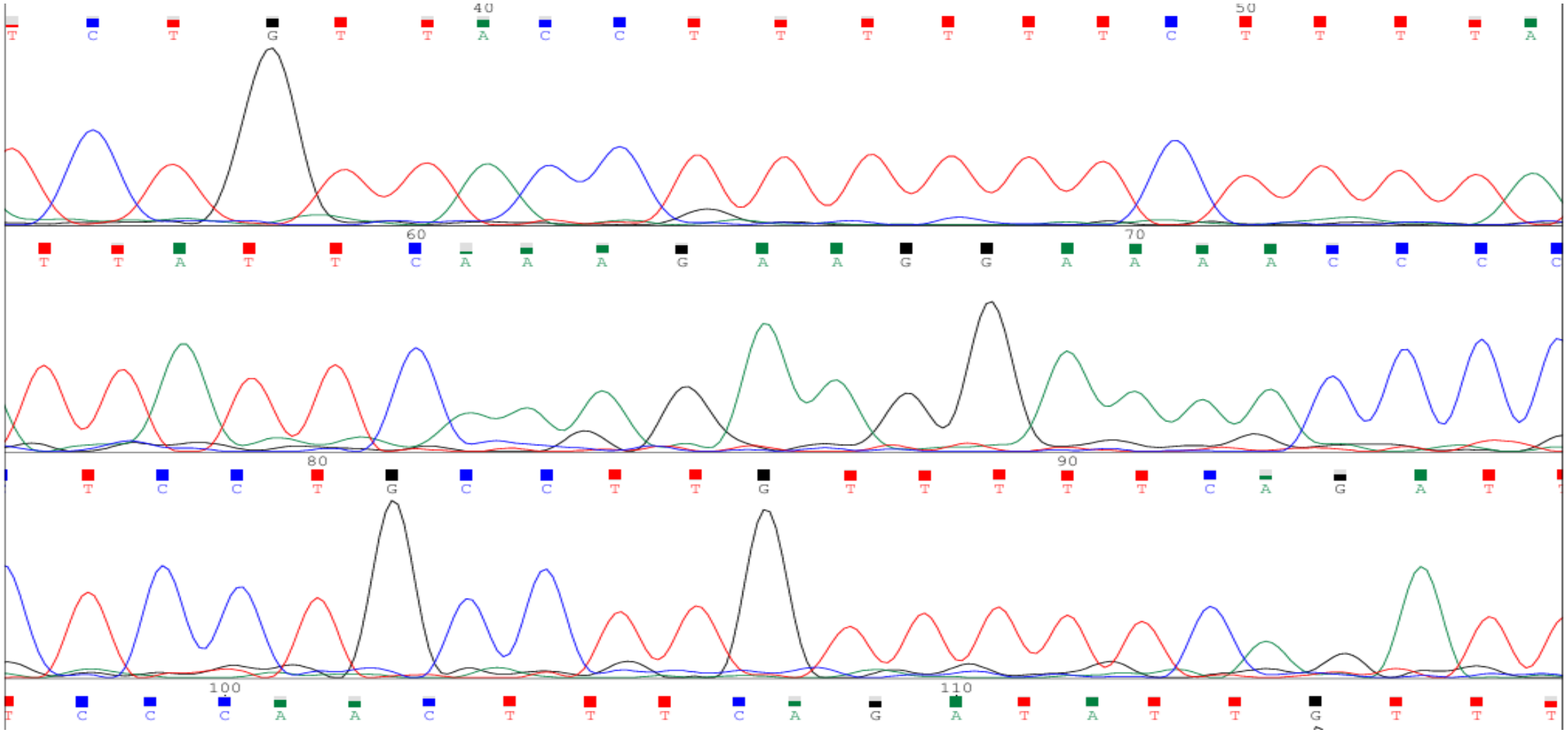
- Sequencing data were accessed using Chromas software.
- Comparison of both sequences (i.e., query and reference sequence) was performed by using the global alignment option in NCBI Blast.
- The sequencing findings indicate the absence of any mutation in coding exons 3 in both families (PKID0139 and PKID0140).

File: PKID0139Azba_MLC1Exon3-F_TSS20250715-0871-01673-BSND_E10.ab1
Signal G:2767 A:2710 T:2382 C:1935
Lane: 72 Base spacing: 16.51
Page 3 of 4

Run Ended: Jul 15, 2025, 15:25:00
Sample: 10E_E-.....PKID0139Azba_MLC1Exon3-F_TSS2025071
390 bases in 4746 scans



Chromatogram of family PKID0139



Chromatogram of family PKID0139

CONCLUSION

- It is concluded that coding exon 3 of *MLC1* gene was not responsible for causing ID in identified families.

RECOMMENATIONS

- Enhance Genetic Screening.
- Conduct awareness campaigns to educate the public.
- Increase genetic counseling for intellectually disabled families.

SIGNIFICANCE

- Advancing genetic understanding of the non-syndromic intellectual disability (NSID).
- Crucial for understanding autosomal recessive inheritance often overlooked in intellectual disability
- Potential genetic marker for NSID
- Improves diagnostic accuracy

ACKNOWLEDGMENT

Allah Almighty

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Prof. Dr. Shagufta Naz

My parents and lab fellows